

BRIGHTSPACE FILES DUMP

- https://drive.google.com/drive/folders/1g7XvQzT-gTlyElcvz35v-GdUoG406HTc?usp=drive_link

HELPFUL INFORMATION

- Watch any of following if needed:
 - Normalization
 - <https://www.youtube.com/watch?v=ABwD8lYByfk>
 - <https://www.youtube.com/watch?v=J-drts33N8g>
 - https://www.youtube.com/watch?v=GFQaEYEc8_8
 - Foreign Keys:
 - <https://www.youtube.com/watch?v=rFssf37UJw>
 - <https://www.youtube.com/watch?v=5kiMg7GXAsY>
 - ERDs
 - <https://www.youtube.com/watch?v=6uwuNRUUimY>
 - <https://www.youtube.com/watch?v=xsg9BDiwiJE> (Part 1)
 - <https://www.youtube.com/watch?v=hktyW5Lp0Vo> (Part 2)
 - UDTs (The professor also had a lecture on this last class, so please watch that video if you weren't there that day)
 - <https://learn.microsoft.com/en-us/sql/t-sql/statements/create-type-transact-sql?view=sql-server-ver17>
 - Ex:
 - ```
CREATE TYPE ZipCode
FROM CHAR(5)
NOT NULL;
```
      - ```
CREATE TYPE StateCode
FROM CHAR(2)
NOT NULL;
```
 - ```
CREATE TYPE PhoneNumber
FROM VARCHAR(12)
NOT NULL;
```
  - ITVF
    - <https://www.geeksforgeeks.org/sql/inline-table-valued-function-in-sql-server/>
    - <https://www.sqlservercentral.com/articles/creating-and-using-inline-table-valued-functions>
    - <https://www.sqlshack.com/sql-server-inline-table-valued-function/>
  - Create Table
    - [https://www.w3schools.com/sql/sql\\_create\\_table.asp](https://www.w3schools.com/sql/sql_create_table.asp)
    - [https://www.tutorialspoint.com/t\\_sql/t\\_sql\\_create\\_tables.htm](https://www.tutorialspoint.com/t_sql/t_sql_create_tables.htm)

- Create Schemas
  - <https://www.geeksforgeeks.org/sql/create-schema-in-sql-server/>
- Self Reference Table
  - <https://sudhass.medium.com/self-referencing-foreign-key-constraint-in-rdbms-and-self-join-b66186e672f7>
- FOREIGN KEYS
  - [https://www.w3schools.com/sql/sql\\_foreignkey.asp](https://www.w3schools.com/sql/sql_foreignkey.asp)

## **GENERAL TASKS**

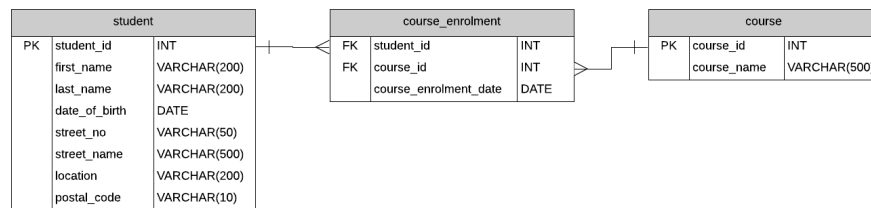
- Run this query first on your SQL Editor:
  - [https://drive.google.com/file/d/1YaGEAC63BsfXwM2yMAqL\\_ejsAMi3qjoC/view?usp=drive\\_link](https://drive.google.com/file/d/1YaGEAC63BsfXwM2yMAqL_ejsAMi3qjoC/view?usp=drive_link)
    - Note: This SQL file creates the PresitgeCars tables, with the bad structure that we need to fix.
- There is also another sql file that shows :
  - [https://drive.google.com/file/d/1e5dgR63LB4UYx6NbTg3hH3muwchdCdcV/view?usp=drive\\_link](https://drive.google.com/file/d/1e5dgR63LB4UYx6NbTg3hH3muwchdCdcV/view?usp=drive_link)
    - Note: This one shows how to create a table properly with constraints. If you want to run this, you need to add this at the top of that file:
      - DROP DATABASE IF EXISTS [PrestigeCars];  
GO  
CREATE DATABASE [PrestigeCars];  
GO  
USE [PrestigeCars];  
GO
- Basically, it gives an example of how to possibly fix one of the tables by using UDTs and constraints for the Country table.
- Please read over the PDFs, and the assignment posted on BrightSpace carefully

## **DESIGN GROUP TASKS**

- Please remember that this is the design, you do not need to write the code. Just document and draw the design of the tables. These notes and drawings will be used for your side of the presentation.
- Task 1
  - Look over the current database made from the .sql file I mentioned above. Look at how the data looks for the following tables:

- Data.Customer
  - Data.Country
  - Data.Make
  - Data.Model
  - Data.Sales
  - Data.SalesDetails
  - Data.Stock
  - Data.PivotTable (do this one last I think)
- While looking over the columns, ask the following questions
  - Do any of the data types look wrong? Should a column be changed to another data type? Do lengths stay consistent so can be specified using a UDT?
  - Does data repeat and can instead be moved to another table?
  - Are NULLs allowed where they shouldn't be?
  - Any relationships missing or need to be created between tables?
  - Rules that need to be enforced? Like length or format?
  - Any parent and child relationships?
  - Do they belong in a business domain (Person, HR, Sales, etc) ?
- Document all this information in a separate google doc
- Task 2:
  - After reviewing all the information, share it with each other and propose a new structure for the tables.
  - Suggested changes will be written in the "Prestige Cars Metadata.xlsx" or use this link:  
<https://docs.google.com/spreadsheets/d/1PAcc0nGSwwvz32DPgEVxwUA4zsY1V8o3/edit?gid=954946963#gid=954946963> to work collaborate
  - Select proper schema names for the database that these tables belong to.
  - Check for normalization:
    - Do tables need to be split to 2 or more tables?
    - Do we need to create lookup tables?
  - Columns:
    - Do they need to be renamed? Do new columns need to be added? Dropped? Datatype changed like make them Udt instead and what does that UDT look like?
    - NULL allowed or not?
    - Any rules that they need to follow? Like if a column is a Price, it can't go lower than 0.
    - Any default values to prevent having NULLs?
    - Do primary keys need to be changed?
    - Should some columns be Foreign keys? Or need to be added?
  - If a column represents a business concept, such as Price, CustomerName, and such it should be a UDT. Define which ones need to be created or used.
- Task 3:

- Now that the design for the tables have been completed, draw them in <https://app.diagrams.net/> or <https://www.lucidchart.com/> (save the file if using draw.io)
  - Note: I think lucidcharts allow for cooperative work, and should be free if you used your gmail.
- <https://www.youtube.com/watch?v=JYZPdU5F2iM> or <https://www.youtube.com/watch?v=RBZtPhZkUZM&t=7s>
- Then draw the final ERD design of the database (This is more of a draft, implementation team will have the full one and should look similar to what design team drew)



## IMPLEMENTATION GROUP TASKS

- Task 1
  - Look through the ERD file and understand what the tables and columns look like.
  - Look at the ERD Reference Data for further details on datatypes, size, and constraints that we will be implementing for our tables.
  - Ask each other any questions to the design team if we don't understand anything
- Task 2
  - Create another database called PrestigeCars\_Project3 as our work environment, this is so we can refer back to the original PrestigeCars if we need to look through it.
  - Create the following in .sql files and putting them in the implementation folder:
    - Creating Schemas
    - Creating UDTs? - don't know if still doing this
    - Creating Tables
    - Creating ITVFs
- Task 3
  - Put it all together, running everything in order and test that these tables works
  - Load some sample data (we can probably just copy and paste some records that already exist in the original PrestigeCars.sql)
  - Test ITVFs work and return data
- Task 4
  - Look at everything we have done so far and verify everything checks out
  - If so then we can rename the database to PrestigeCars and then create .bak file

- Use SSMS to get the final ERD diagram
- Get ready for presentation
- Record video